

cryptography

PREPARED by Sirajudin
CTC-153-26

← Back to all walkthroughs



Cryptography Basics

Learn the basics of cryptography and symmetric encryption.

  45 min  158,627 

[Share your achievement](#) [Save Room](#) [1640 Recommend](#) [Connectivity](#) [Options](#) ▼

Room completed: 100%

Cryptography's ultimate purpose is to ensure secure communication in the presence of adversaries. The term secure includes confidentiality and of the communicated data. Cryptography can be defined as the practice and study of techniques for secure communication and data protection where we expect the presence of adversaries and third parties. In other words, these adversaries should not be able to disclose or alter the contents of the messages.

We have just introduced several new terms, and we need to learn them to understand any text about cryptography. The terms are listed below:

- **Plaintext** is the original, readable message or data before it's encrypted. It can be a document, an image, a multimedia file, or any other binary data.
- **Ciphertext** is the scrambled, unreadable version of the message after encryption. Ideally, we cannot get any information about the original plaintext except its approximate size.
- **Cipher** is an algorithm or method to convert plaintext into ciphertext and back again. A cipher is usually developed by a mathematician.
- **Key** is a string of bits the cipher uses to encrypt or decrypt data. In general, the used cipher is public knowledge; however, the key must remain secret unless it is the public key in asymmetric encryption. We will visit asymmetric encryption in a later task.
- **Encryption** is the process of converting plaintext into ciphertext using a cipher and a key. Unlike the key, the choice of the cipher is disclosed.
- **Decryption** is the reverse process of encryption, converting ciphertext back into plaintext using a cipher and a key. Although the cipher would be public knowledge, recovering the plaintext without knowledge of the key should be impossible (infeasible).

Answer the questions below

What do you call the encrypted plaintext?

ciphertext ✓ Correct Answer

What do you call the process that returns the plaintext?

decryption ✓ Correct Answer

Categories of encryption

symmetric and asymmetric

Symmetric encryption is a method in which the same key is used for both encryption and decryption. Consequently, this key must remain secure and never be disclosed to anyone except the intended party.

Asymmetric encryption is a method that uses two different keys: a public key for encryption and a private key for decryption.

Answer the questions below

What's $1001 \oplus 1010$?

0011

✓ Correct Answer

What's $118613842\%9091$?

3565

✓ Correct Answer

What's $60\%12$?

0

✓ Correct Answer